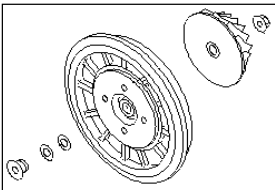


|                                                                                     |                   |  |              |             |
|-------------------------------------------------------------------------------------|-------------------|--|--------------|-------------|
| <b>DISASSEMBLY CHECK SHEET</b><br><b>TURBO CHARGER</b><br>KOMATSU ENGINE ALL SERIES | JOB NUMBER        |  | DOCUMENT NO  | QR02-SA-003 |
|                                                                                     | ENGINE MODEL      |  | PUBLIC.DATE  | 20/08/2008  |
|                                                                                     | ENGINE SERIAL NO. |  | REVISION NO. | 03          |
|                                                                                     | MACHINE MODEL     |  | PAGE         | 1 of 3      |

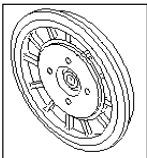
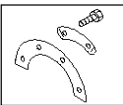
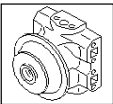
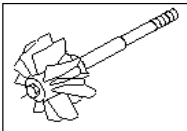
|         |        |  |                |  |
|---------|--------|--|----------------|--|
| PROCESS | START  |  | DISASSEMBLY BY |  |
|         | FINISH |  |                |  |

**Note :**

- Before disassembling the turbocharger assembly , measure the clearance between the blower impeller and the housing by using filler gauge.

| NO. | PARTS TO REMOVE OR INSPECT | CONDITIONS TO BE CHECKED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | REUSE | SALVAGE | REPLACE | SKETCH DRAWING OR PHOTOES,                                                                                                                                                                                                                                     | REMARKS |
|-----|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1   | Blower Housing             | <ul style="list-style-type: none"> <li>When removing the blower housing, take care not to damage the blower impeller.</li> <li>Remove the nut and V-band clamp<br/>Remove blower housing</li> <li>Before removing the blower impeller, measure the end play and the radial play of the rotor assembly in the following manner and keep the measurement on record               <ol style="list-style-type: none"> <li>Measurement of the end play                   <ul style="list-style-type: none"> <li>Set dial gauge in position<br/>Read the dial gauge indicator when the rotor assembly is moved in the axial direction.</li> </ul> </li> <li>Measurement of the radial play                   <ul style="list-style-type: none"> <li>Set dial gauge in position<br/>Read the dial gauge indicator when the rotor assembly is moved in the radial direction</li> </ul> </li> </ol> </li> </ul> |       |         |         |    |         |
| 2   | Blower Impeller            | <ol style="list-style-type: none"> <li>Hold on the end of turbine rotor with the socket wrench or the others special tools</li> <li>Loosen and remove nut</li> <li>Extract the blower impeller from the turbine rotor in the following manner               <ul style="list-style-type: none"> <li>Immerse the rotor assembly in the oil that is heated at 140 to 160°C for 5 to 20 minutes or with hot air gun</li> <li>Extract blower impeller from the turbine rotor</li> <li>When extracting the blower impeller, take care not to apply an excessive force to the impeller</li> </ul> </li> </ol>                                                                                                                                                                                                                                                                                                 |       |         |         |                                                                                                                                                                            |         |

|                                                                              |                   |  |              |             |
|------------------------------------------------------------------------------|-------------------|--|--------------|-------------|
| DISASSEMBLY CHECK SHEET<br><b>TURBO CHARGER</b><br>KOMATSU ENGINE ALL SERIES | JOB NUMBER        |  | DOCUMENT NO  | QR02-SA-003 |
|                                                                              | ENGINE MODEL      |  | PUBLIC.DATE  | 20/08/2008  |
|                                                                              | ENGINE SERIAL NO. |  | REVISION NO. | 03          |
|                                                                              | MACHINE MODEL     |  | PAGE         | 2 of 3      |

| NO. | PARTS TO REMOVE OR INSPECT | CONDITIONS TO BE CHECKED                                                                                                                         | REUSE | SALVAGE | REPLACE | SKETCH DRAWING OR PHOTOES                                                             | REMARKS |
|-----|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------|---------|---------------------------------------------------------------------------------------|---------|
| 3   | Insert (Back Plate)        | 1. Straighten the lock washer and remove bolts<br><br>2. Pull out the back plate from the center housing and turbine shaft.                      |       |         |         |    |         |
| 4   | Flinger                    | 1. Pull out flinger from the insert (back plate)<br><br>2. Remove seal ring from the flinger                                                     |       |         |         |    |         |
| 5   | Thrust Bearing             | <ul style="list-style-type: none"> <li>Remove thrust bearing</li> </ul>                                                                          |       |         |         |    |         |
| 6   | Thrust Collar              | <ul style="list-style-type: none"> <li>Remove thrust collar</li> </ul>                                                                           |       |         |         |    |         |
| 7   | Journal Bearing            | <ul style="list-style-type: none"> <li>Remove journal bearing</li> </ul>                                                                         |       |         |         |  |         |
| 8   | Bolt Turbine               | <ul style="list-style-type: none"> <li>Straighten the lock washer, remove mounting bolt and then remove lock plate.</li> </ul>                   |       |         |         |  |         |
| 9   | Center Housing             | <ul style="list-style-type: none"> <li>Pull out center housing from the turbine rotor and the housing turbine with special tools</li> </ul>      |       |         |         |  |         |
| 10  | Shield                     | <ul style="list-style-type: none"> <li>Remove shield from turbine housing.</li> </ul>                                                            |       |         |         |  |         |
| 11  | Turbine Rotor              | <ul style="list-style-type: none"> <li>Remove turbine rotor from turbine housing and remove mounting bolt and then remove Lock plate.</li> </ul> |       |         |         |  |         |

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|----------------------|
| INSPECTION COMMENT : |
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|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| INSPECTION BY                                                                                  | APPROVED BY                                                                                              |
| <div style="border-top: 1px solid black; width: 100%; margin-bottom: 5px;"></div> ( MECHANIC ) | <div style="border-top: 1px solid black; width: 100%; margin-bottom: 5px;"></div> PROD. SUPERVISOR (SPV) |

|                                                                                     |                   |  |              |             |
|-------------------------------------------------------------------------------------|-------------------|--|--------------|-------------|
| <b>DISASSEMBLY CHECK SHEET</b><br><b>TURBO CHARGER</b><br>KOMATSU ENGINE ALL SERIES | JOB NUMBER        |  | DOCUMENT NO  | QR02-SA-003 |
|                                                                                     | ENGINE MODEL      |  | PUBLIC.DATE  | 20/08/2008  |
|                                                                                     | ENGINE SERIAL NO. |  | REVISION NO. | 03          |
|                                                                                     | MACHINE MODEL     |  | PAGE         | 3 of 3      |